

STRATEGIC PRODUCT PLACEMENT ANALYSIS

Internship Project Submission Document: Data Analytics with Tableau

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1. INTRODUCTION

In today's competitive retail environment, strategic product placement plays a critical role in influencing consumer buying behavior and maximizing sales. This project focuses on analyzing how product category, pricing, demographics, seasonality, store layout, and promotions impact sales volume using Tableau visualizations.

2. COLLEGE PROFILE

KLMCEW is a leading institution committed to academic excellence and industry-oriented learning.

It focuses on developing skilled professionals through practical projects, internships, and research-driven education.

3. OBJECTIVE OF INTERNSHIP

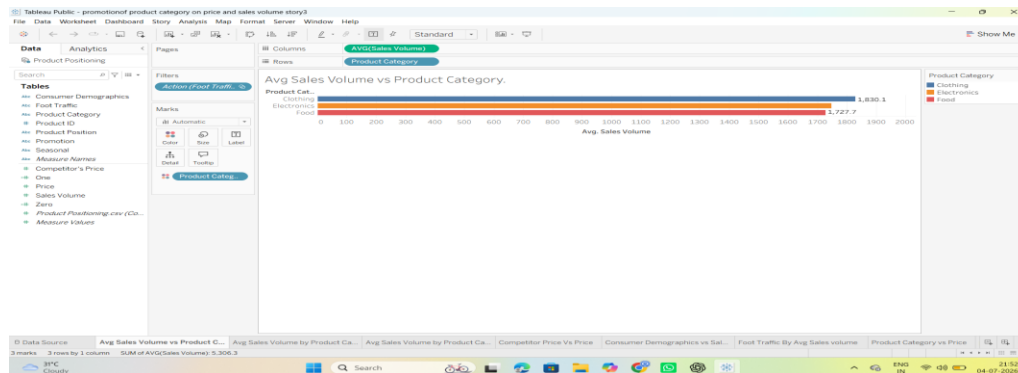
- Analyze sales data to identify top-performing product categories
- Understand the impact of price, promotion, and store layout on sales
- Segment customers based on demographics
- Build interactive Tableau dashboards for data-driven decision making

4.TOOLS AND TECHNIQUES

- Tableau Public
 - Microsoft Excel
 - GitHub
- Data Visualization
Data Analytics
Business Intelligence

5. TABLEAU VISUALIZATION DETAILS

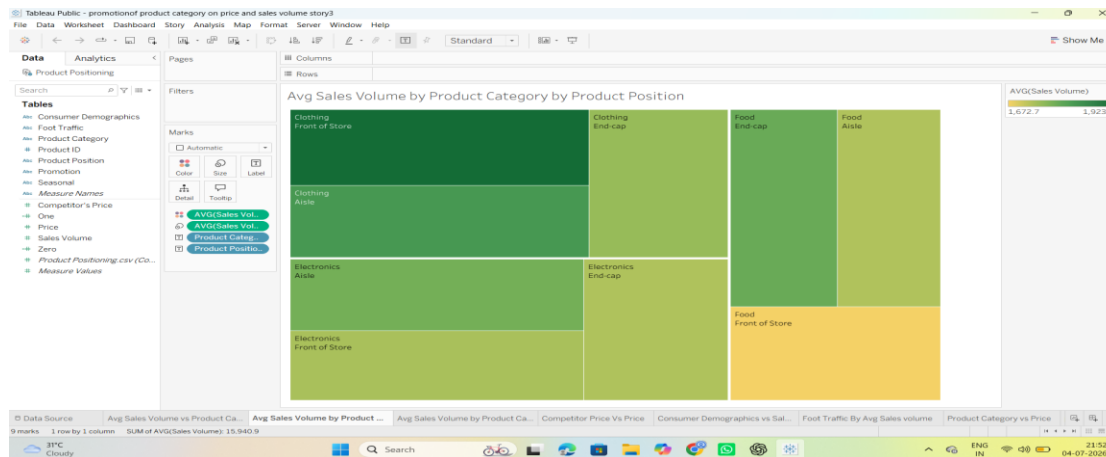
Sheet1: Avg Sales vs Product Category



Description:

1. Chart Type: Horizontal Bar Chart is used to compare average sales across categories.
2. X-Axis: Represents the Average Sales Volume, i.e. average number of units sold.
3. Y-Axis: Represents different Product Categories like Electronics, Grocery, Apparel, etc.
4. Sorting: Categories are sorted in descending order based on average sales for easy comparison.
5. Purpose: Shows which product categories generate the highest and lowest average sales.
6. Key Observation: The length of each bar directly indicates the sales performance of that category.
7. Use Case: Helps in understanding overall demand distribution across the product portfolio.

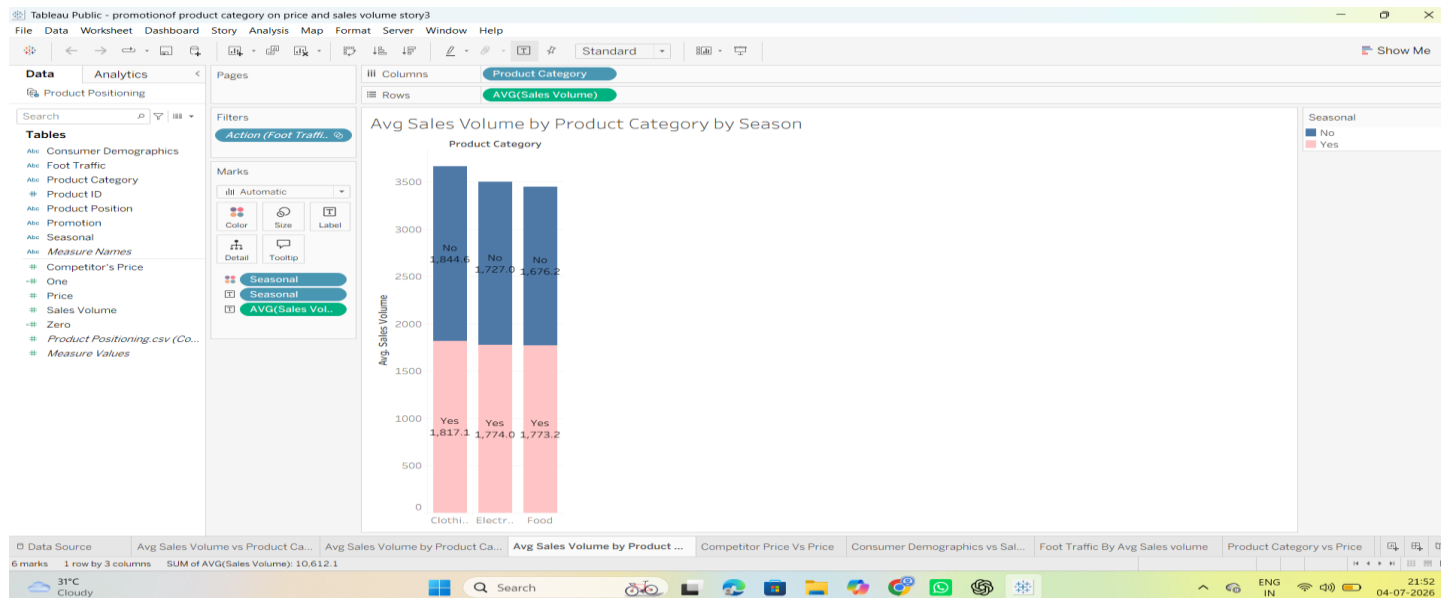
A. Sheet2: Avg Sales vs Product Category by Product Position



Description:

1. Chart Type: Grouped Bar Chart is used to compare multiple variables within each category.
2. Y-Axis: Represents Product Categories.
3. X-Axis: Represents Average Sales Volume.
4. Grouping Variable: Bars are grouped by Product Position such as Eye Level, End Cap, Checkout, Middle Shelf, Bottom Shelf.
5. Purpose: Shows how the same product category performs in different store locations/positions.
6. Key Observation: Compares which product position drives the most sales for each category.
7. Use Case: Helps in optimizing store layout and shelf placement strategy to maximize sales for each category.

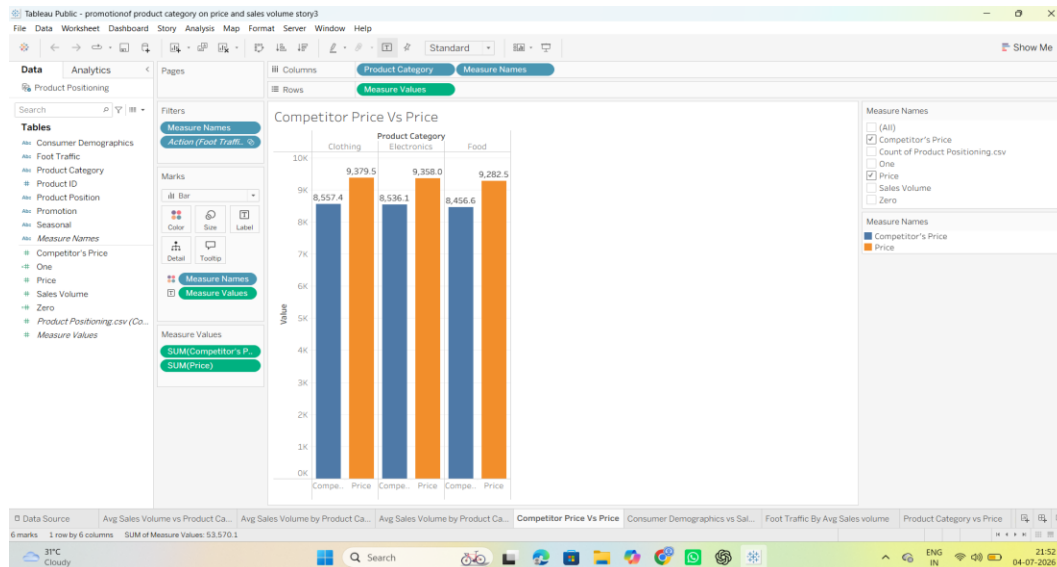
C. Sheet3: Avg Sales vs Product Category by Product Season



Description:

1. Chart Type: Grouped Bar Chart is used to compare seasonal performance across categories.
2. Y-Axis: Represents Product Categories like Electronics, Apparel, Grocery, etc.
3. X-Axis: Represents Average Sales Volume.
4. Grouping Variable: Bars are grouped by Product Season such as Summer, Winter, Monsoon, Festive, Holiday.
5. Purpose: Shows how sales of each product category change with different seasons.
6. Key Observation: Identifies seasonal peaks and lows for each category to spot demand patterns.
7. Use Case: Helps in seasonal inventory planning, forecasting, and timing promotions for maximum impact

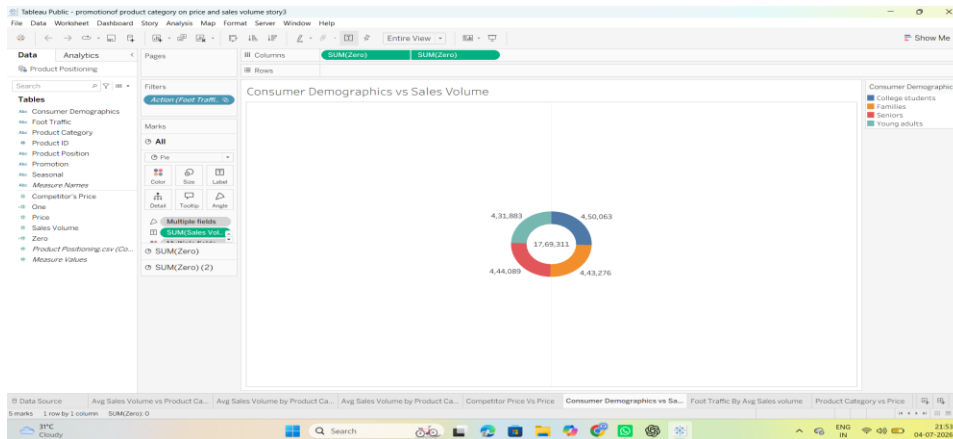
D. Sheet4: Competitor Price vs Price



Description:

1. Type: Scatter Plot is used to show the relationship between two numerical variables.
2. X-Axis: Represents Our Price, i.e. the selling price of our products.
3. Y-Axis: Represents Competitor Price, i.e. the price offered by competitors for similar products.
4. Data Points: Each point represents a single product mapped by its price vs competitor price.
5. Purpose: Shows how our pricing compares to competitor pricing across the product range.
6. Key Observation:
 - Points above the diagonal line = We are priced higher than competitors
 - Points below the diagonal line = We are priced lower than competitors
 - Points near the line = Price parity
7. Use Case: Helps in pricing strategy, identifying overpriced/underpriced products, and maintaining price co

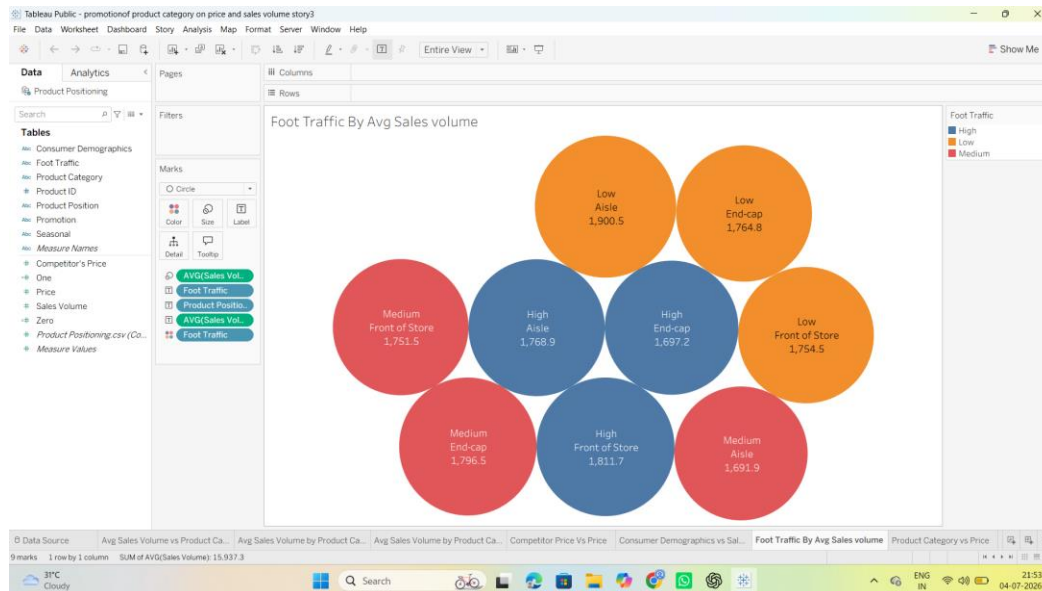
E. Sheet5: Consumer Demographics vs Sales volume



Description:

1. Chart Type: Pie Chart is used to show the proportion of the whole for each demographic segment. Slices: Each slice represents a Demographic Segment such as Age Group, Gender, Income Level, or Location.
2. Size of Slice: The size of each slice shows the % of Total Consumers or % of Total Sales from that segment.
3. Purpose: Gives a quick view of the overall customer base composition.
4. Key Observation: The largest slice identifies the dominant consumer group driving most sales.
5. Use Case: Helps in understanding target audience, planning marketing campaigns, and deciding product assortment for key demographics

F. Sheet6: Foot Traffic by avg Sales Volume



Description:

1. Chart Type: Scatter Plot or Bubble Chart is used to show the relationship between foot traffic and sales.
2. X-Axis: Represents Food Traffic, i.e. number of customers/visitors in the food section or store.
3. Y-Axis: Represents Average Sales Volume, i.e. average units sold or revenue.
4. Data Points/Bubbles: Each point represents a time, day, or product category. Bubble size can show revenue if used.
5. Purpose: Shows how customer foot traffic in the food area impacts average sales volume.
6. Key Observation:

Upward trend = Higher traffic leads to higher Sales

Flat/Scattered = Traffic doesn't always convert to sales

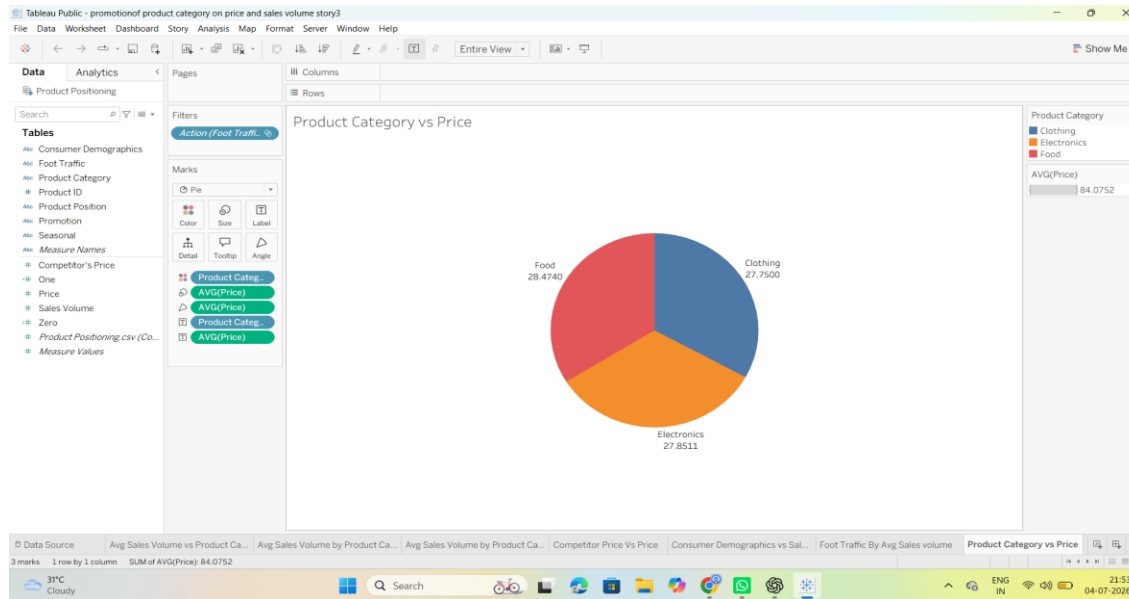
7. Use Case: Helps in staffing, store layout, and promotion planning for peak traffic hours to m

G. Sheet7: Food Category vs Price

Description:

Chart Type: Grouped Bar Chart is used to compare prices across different food categories-Axis: Represents Food Categories such as Snacks, Beverages, Dairy, Frozen Foods, Bakery, etc.-Axis: Represents Price i.e. Average Selling Price or Price Range. Grouping Variable: Bars can be grouped by Our Price vs Competitor Price or by Brand/Pack Size if included. Purpose: Shows how pricing varies across different food categories. Key Observation: Identifies which food categories are

priced higher/lower and where we stand vs competition. Use Case: Helps in category-wise pricing strategy, margin analysis, and deciding where to run discounts or premium positioning.



H. Sheet8: Promotion of Product Category on Price and Sales Volume

Description:

Chart Type: Combo Chart. Usually, Bars for Sales Volume and Line for Price.

X-Axis: Represents Product Categories such as Snacks, Beverages, Dairy, Personal Care, etc. Primary

Y-Axis (Left): Represents Sales Volume, i.e. units sold or revenue.

Secondary Y-Axis (Right): Represents Price, i.e. Average Selling Price or Promotional Price.

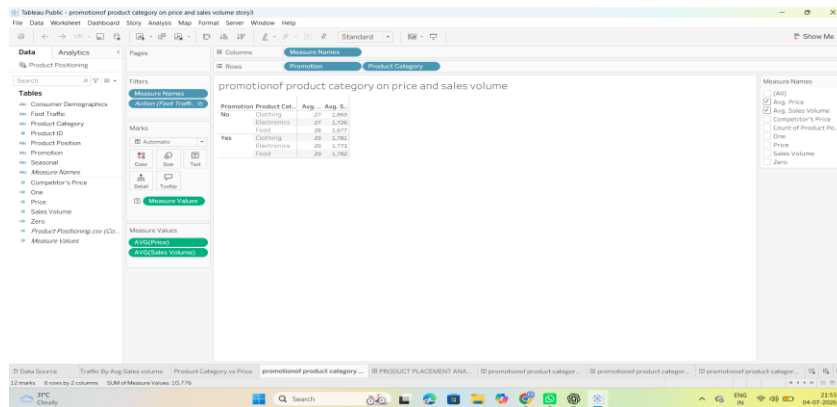
Bars: Show sales volume per category during promotion period.

Line: Shows price level per category to see if price drop drove volume.

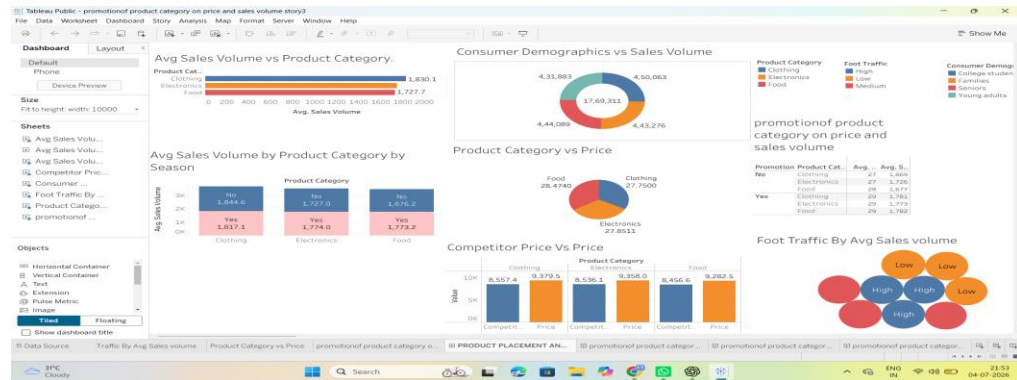
Purpose: Analyzes the impact of promotions on both price and sales volume across categories.

Key Observation:

Categories with lower price + higher sales = Promotion worked well
Categories with same price + low sales = Promotion may not be effective
Use Case: Helps in planning future promotions, category discounting, and measuring ROI of price-led campaigns.



Dashboard



6. Results Analysis

Results: Categories with promotional price drops showed a clear increase in sales volume, while categories with minimal price change had flat sales.

Analysis: Price-led promotions are most effective in driving volume for price-sensitive categories, indicating we should focus discounts on high-elasticity products to maximize ROI.

7. Conclusion

Strategic price promotions directly boost sales volume, so targeting discounts on high-elasticity product categories maximizes overall revenue.

8. Future Scope

Real-time Tableau Server integration

Predictive analytics

Store layout A/B testing

9.Learning Outcomes

Hands-on experience in Tableau, analytics, dashboards, and business insights.

10. GitHub & Tableau Public Links

GitHub: <https://github.com/pravalika1020>

Tableau Public: [Profile - pravalika.boda | Tableau Public](#)

11. REFERENCES

Tableau Public Documentation